Student Marks Analysis

B.Tech Python Programming – Activity-Based Learning

Python source code only. Save or copy the program into a .py file and run it with Python 3.

```
marks = []
subjects = ["English", "Mathematics", "Physics", "Programming", "Chemistry"]

for subject in subjects:
    while True:
        try:
            mark = float(input(f"Enter {subject} marks (0-100): "))
            if 0 <= mark <= 100:
                marks.append(mark)
                break
            print("Marks must be between 0 and 100.")
        except ValueError:
            print("Please enter a valid number.")

total = sum(marks)
percentage = total / len(subjects)

if any(mark < 35 for mark in marks):
    result = "Fail"
elif percentage >= 75:
    result = "Distinction"
elif percentage >= 60:
    result = "First Class"
elif percentage >= 50:
    result = "Second Class"
else:
    result = "Pass"

print("\n----- STUDENT RESULT -----")
for subject, mark in zip(subjects, marks):
    print(f"{subject}: {mark:g}")
print(f"Total: {total:g}/500")
print(f"Percentage: {percentage:.2f}%")
print("Result:", result)
```